Brian Tam

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Education

Rensselaer Polytechnic Institute Troy, NY

Bachelor of Science in Computer Science and Information Technology & Web Science *Aug. 2024 - May 2028*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Architecture, Theory of Computation, Artificial Intelligence

Experience

Software Engineer Intern May 2026 – Aug. 2026

Isaac Morris Limited *New York, NY*

- Engineered a changelog platform processing 1M+ database events daily, saving 5+ hours/week of manual investigation through event deduplication and strict Row-Level Security.
- Built a React document designer for 500+ employees, replacing a legacy system with automatic overflow handling, live annotations, and parallel processing for an estimated 5× speedup in document generation.
- Developed an automated PDF and image-processing pipeline that enables designers to download thousands of images in seconds, accelerating new clothing design and saving 7+ hours/week through concurrent batch processing.

Undergraduate Teaching Assistant Aug. 2025 – Present

Rensselaer Polytechnic Institute *Troy, NY*

- Facilitate weekly laboratory sessions for 50+ CS1 students, and translate complex concepts like recursion and data structures into approachable Python exercises.
- Lead high-volume office hours, actively diagnosing diverse logic errors and teaching industry standard debugging strategies.
- Developed targeted review materials and led exam preparation sessions, contributing to a 5% increase in average exam scores after each assessment.

Projects

Litter-ally Trash | Best Hardware Hack, Cornell BigRed//Hacks | *Python, OpenAI CLIP, CUDA* Oct. 2026

- Won Best Hardware Hack and placed top 4 of 58 teams at Cornell BigRed//Hacks for an AI-powered trash bin on wheels designed to make waste disposal more accessible.
- Developed an OpenAI CLIP ViT-L/14 computer vision system to distinguish trash from recyclables in a live video feed and automatically open the corresponding bin lid.
- Integrated hand-gesture, voice, and keyboard control with a Raspberry Pi and Arduino-powered Mecanum drive base, enabling accessible, omnidirectional steering from phones and laptops in just 36 hours.

ByteBite | **Best Data Science Hack, HackRPI 2025** | *React, JavaScript, Node.js, GitHub, Render* Nov. 2025

- Built an AI-powered dining recommendation platform using normalized Sodexo data and real-time ranking logic.
- Engineered a schema-inference parser for 20+ dynamic menu formats and integrated a Gemini recommendation engine to match daily inventory against complex user dietary constraints.
- Recognized by a Google Cloud engineer as one of the standout projects out of 400+ participants, highlighted for its scalability and real-world impact.

The Thread | *React, TypeScript, Node.js, MongoDB, Axios, Vercel* Jan. 2025 – May 2025

- Built a React/TypeScript platform through the Rensselaer Center for Open Source designed to support editorial content for thousands of undergraduate students.
- Implemented core React components mapped to a MongoDB + AWS S3 backend, optimizing the pipeline to dynamically render high-resolution editorial media and articles with minimal latency.
- Collaborated with a 6-person team using agile methodologies; enforced code quality through rigorous peer reviews and automated CI/CD pipelines to ensure constant uptime.

Technical Skills

- **Languages**: C, C++, Python, Java, TypeScript, JavaScript, SQL, PHP, MIPS
- **Frameworks & Web**: React, Next.js, Node.js, Express.js, Tailwind CSS, HTML/CSS, REST APIs
- **Databases & Cloud**: PostgreSQL, MySQL, MongoDB, Supabase, Snowflake, AWS S3, Azure, DigitalOcean
- **Tools & AI**: Git, Docker, Linux/Unix, CI/CD, scikit-learn, RAG, Semantic Search, LLM APIs